Report for research paper

ON

Advantages and Disadvantages of Modern Language Models

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING
In
COMPUTER SCIENCE AND ENGINEERING

Submitted by:

Bhuvain Jhamb – 2210990228
Samiksha - 2210992587
Daksh Chawla - 2210990243

Supervised By:

Monika Aggarwal

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
CHITKARA UNIVERSITY
CHANDIGARH-PATIALA NATIONAL HIGHWAY, RAJPURA, PUNJAB, INDIA

CONTENTS

DECLARATION

We hereby declare that the project work titled, “Advantages and Disadvantages of Modern Language Models” submitted as part of Bachelor’s degree in CSE, at Chitkara University, Punjab, is an authentic record of our own work carried out under the supervision of the Faculty Mentor.

Signature(s):

Bhuvain Jhamb

Samiksha

Daksh Chawla

ACKNOWLEDGEMENT

The authors acknowledge Chitkara University for the institutional support and computational resources provided, which enabled the successful completion of this research project. The support of the Department of Computer Science and Engineering in facilitating the academic environment necessary for this work is gratefully recognised.

ABSTRACT

The improvement of model performance during benchmarking, due to increases in the number of parameters and tokens within a sequence, is quite noticeable. However, in practice, large language models (LLMs) exhibit several drawbacks, including inefficient memory bandwidth usage, prolonged inference periods, energy wastage, and poor scalability. Merely increasing the number of parameters is therefore not a sufficient solution; rather, a holistic approach accounting for physical constraints, cost efficiency, and sustainability is required.

This report compares two primary LLM architectural paradigms — Retrieval-Augmented Generation (RAG) approaches and extended-context transformers — using a unified Token Economics metric. The Token Economics framework measures LLM design effectiveness through three principal indicators: token throughput speed, memory bandwidth costs, and energy-normalised inference time. Experiments are conducted on LLMs ranging from 7B to 70B parameters with context windows spanning 256 to 1 million tokens, evaluated across static reasoning and dynamic knowledge generation workloads.

The findings reveal that retrieval-augmented models are more cost-effective for dynamic knowledge-based tasks and demonstrate superior efficiency under bandwidth-constrained conditions. Extended-context transformers, while powerful for static reasoning tasks, face significant limitations arising from the scaling of the attention mechanism and KV-cache management. Key infrastructure constraints — including interconnect bandwidth, rack power density, and thermal generation — are identified as primary obstacles to scaling.

Index Terms — Token Economics, Retrieval-Augmented Generation, Memory Bound Computing, Infrastructure Scalability, Hardware Constraints.

1. Introduction

The artificial intelligence industry has achieved remarkable progress since its inception. The development of large language models (LLMs) has fundamentally transformed the capacity of computers to process, comprehend, generate, and utilise natural language. LLMs have grown substantially, from models with a few million parameters in 2017, when the transformer architecture was first introduced, to those containing billions — and in some cases more than a trillion — parameters in recent years [1]. These models are capable of solving complex problems, generating code, composing narratives, and conducting scientific research. The progression from GPT-3's 175 billion parameters to contemporary models of even greater scale is a testament to the rapid technological advancement in this domain [2]. LLMs are now capable of reading entire books, sustaining extended conversations, and reasoning over intricate problems — tasks that were previously unattainable. Consequently, these models have drawn significant attention from professionals in the fields of healthcare, education, research, and enterprise.

Despite this progress, the paradigm of aggressive parameter scaling faces mounting physical and economic pressures. As models grow in complexity — with increasing parameter counts and ever-expanding context windows — practical challenges emerge in the form of inadequate memory bandwidth, slow inference speeds, high power consumption, and limited system scalability [3]. Recent research indicates that real-world deployment constraints cannot be bypassed by simply increasing parameters [4]. Furthermore, the rapid commercialisation of autonomous AI agents underscores the importance of architectural decisions for broader digital ecosystems. Medical diagnostic applications, AI-driven e-learning platforms, and enterprise decision-making agents all depend critically on their underlying architecture in terms of efficiency, stability, and environmental footprint [5, 6].

Two principal architectural paradigms have emerged as candidates for addressing these challenges: Retrieval-Augmented Generation (RAG) models, which access external knowledge bases to reduce intrinsic model size, and extended-context transformers, which expand internal architecture to improve in-context reasoning performance [7, 8]. This report addresses the critical need for a rigorous, unified assessment of these architectural choices. Grounded in the Token Economics framework, the analysis evaluates the efficiency, scalability, and sustainability of each approach.

2. System Requirements

2.1 Hardware and Infrastructure Requirements

Upon implementing LLMs in production environments, numerous challenges arise that are not apparent in controlled experimental settings. Pope et al. [3] demonstrated that memory bandwidth — rather than raw computational capacity — is the dominant cost factor when executing large-scale models. Patterson et al. [4] further estimated that the carbon footprint of training a state-of-the-art model is comparable to the emissions generated by transcontinental air travel undertaken multiple times. Hardware constraints including available memory capacity, data transfer rates, power consumption, and thermal management significantly influence both the practical deployment and architectural design of LLMs.

For the deployment of 70B-parameter extended-context models, each server rack requires approximately 8 GPUs consuming between 40 and 60 kW of electrical power — far exceeding the conventional infrastructure power budget of 20–25 kW per rack. High-density GPU configurations also generate substantial heat within confined spaces, necessitating specialised liquid cooling systems. These requirements increase capital expenditure and restrict the geographic availability of deployment sites.

2.2 Software and Architectural Requirements

Two principal architectural paradigms are considered for LLM deployment. The first is Retrieval-Augmented Generation (RAG), in which external knowledge bases are queried at inference time to augment generation with current, domain-specific information, enabling the use of smaller base models. The second is the extended-context transformer architecture, which expands internal capacity to process longer token sequences without external retrieval.

Both paradigms require specific software stacks, GPU drivers, and inference engines for deployment. Three experimental deployment configurations are examined in this study: (1) high-performance cloud servers equipped with NVIDIA A100/H100 GPUs and high-throughput interconnects; (2) edge deployment hardware with constrained memory and lower computational capacity; and (3) serverless deployment environments characterised by infrequent model activation and cold-start latency requirements.

3. Methodology

3.1 Evaluation Framework: Token Economics

This study introduces the Token Economics framework as a comprehensive method for assessing LLM efficiency under real-world deployment constraints. The framework evaluates LLM performance through the lens of token processing, examining concurrent token handling capacity, inter-memory data transfer energy, and per-token energy consumption. The fundamental computational unit — the token — serves as the central metric for assessing model efficiency, with all costs considered per token generated.

The five cost dimensions of the Token Economics Framework are as follows: (1) Token Processing Cost — computational effort per token, expressed in FLOPS; (2) Memory Transfer Cost — the volume of data transferred across memory levels per token; (3) Context Management Cost — the memory and computational overhead of maintaining the sequence during generation; (4) Retrieval Cost (applicable to RAG models) — the cost associated with querying and integrating external knowledge bases; and (5) Energy Cost — the total energy consumed per token produced. This multi-dimensional framework enables precise identification of the strengths and weaknesses of each architectural design, beyond conventional performance-only benchmarks.

3.2 Experimental Design

Comparative experiments evaluate architectural choices across multiple axes. Model scales of 7B, 13B, 30B, and 70B parameters are examined, encompassing edge-deployment models (7B–13B) and data-centre-class models (30B–70B). Context window sizes range from 4K to 1M tokens. Two primary workload categories are considered: static reasoning — encompassing logical, mathematical, and programming tasks in which all required information is present in the query — and dynamic knowledge tasks — requiring access to current or domain-specific information not covered in the training dataset.

RAG experiments vary the retrieval corpus size (1 million to 100 million documents), retrieval system type (dense vs. sparse indexing), and the number of retrieved documents per query (top-5 to top-100). Three deployment environments are simulated: large-scale cloud servers with NVIDIA A100/H100 GPUs and high-bandwidth interconnects; memory-constrained edge hardware; and serverless environments requiring cold-start model initialisation.

3.3 Measurement Methodology

Experimental measurements are organised under two primary categories. Performance metrics include task accuracy on established benchmarks such as MMLU and HellaSwag, output quality assessed through human and automated evaluation, and task completion time across varying sequence lengths and batch sizes. Efficiency metrics encompass memory utilisation measurements — including peak memory consumption, KV-cache capacity, and memory bandwidth usage — as well as energy consumption readings obtained using hardware power meters.

3.4 Analysis Methods

Statistical analyses employ generalised linear mixed models to account for repeated measures across different models and workload configurations, enabling comprehensive evaluation of architectural effects while controlling for confounding variables. Cost-benefit analysis examines the balance between performance gains and operational costs, considering both upfront infrastructure investment and ongoing operational expenditure. Scaling analyses investigate how costs and benefits evolve in relation to model size, context window length, and deployment scale.

3.5 IO-Complexity Metrics and AI Engagement Modalities

To incorporate FlashAttention into the Token Economics framework, IO-complexity evaluation is introduced as a supplement to FLOP counting under the Memory Transfer Cost dimension. Standard attention mechanisms require $\Theta(Nd + N^2)$ high-bandwidth memory (HBM) accesses, whereas FlashAttention reduces this complexity to $\Theta(N^2d^2M^{-1})$, where M denotes on-chip SRAM capacity. This metric is measured experimentally across varying context lengths to quantify memory transfer savings per token.

Two distinct modalities of AI–human interaction are also examined. Active Engagement involves direct collaboration, as exemplified by the GitHub Copilot experiment [25] in which 95 professional developers completed an HTTP-server assignment, with performance measured from repository creation to the first commit passing 12 automated tests. Passive Exposure involves indirect contact with AI-generated output, as studied through the dynamic Alternate Uses Task experiment [28] conducted with 844 participants across 48 countries.

4. Software Design

4.1 Evolution of Large Language Models

The original transformer model, introduced by Vaswani et al. [1], employs a self-attention mechanism that enables parallel processing of input sequences — a significant advantage over the sequential computation required by recurrent architectures. The attention mechanism allows the model to weight the relevance of any element in a sequence relative to any other, providing superior contextual awareness. Subsequent developments, including BERT, GPT-2, and GPT-3, demonstrated the capacity of scaled transformer models to acquire novel capabilities beyond those explicitly present in their pre-training objectives [2]. Kaplan et al. [9] established empirical scaling laws confirming that increasing model size and training compute consistently yields performance improvements.

A persistent challenge of the transformer architecture is the quadratic growth of the attention mechanism with respect to sequence length, making inference computationally expensive for very long inputs. Architectural innovations including Transformer-XL [10], Longformer [11], and Infini-attention [12] were developed to address this limitation, enabling the processing of sequences spanning thousands to millions of tokens.

4.2 Retrieval-Augmented Generation Architecture

The Retrieval-Augmented Generation (RAG) architecture departs from the paradigm of encoding all knowledge within model parameters. Instead, RAG models access relevant information from external knowledge bases at inference time [7]. This approach confers several structural advantages: it permits the use of smaller base models, reduces the computational cost of knowledge updates, and enables access to information that postdates the model's training cutoff. Guu et al. [13] demonstrated that integrating retrieval techniques into the training process of language models substantially enhances task performance while requiring fewer parameters. The REALM architecture and related models confirmed that embedding knowledge acquisition within the training procedure improves accuracy on question-answering and knowledge-intensive tasks.

Lewis et al. [14] further established that RAG-based models can successfully complete a wide range of tasks using significantly fewer parameters than their fully parametric counterparts. The modular architecture of RAG systems also simplifies adaptation to specific application domains, reducing the computational resources required for fine-tuning. Retrieval-based systems such as SciSpace and Elicit demonstrate markedly lower hallucination rates compared to purely generative models [22], making them particularly suitable for applications requiring factual accuracy.

4.3 Extended-Context Transformer Architecture

Extended-context transformers address the limitation of fixed-length context windows by increasing internal capacity to process longer sequences without external retrieval. Transformer-XL [10] applies recurrence at the segment level, enabling the capture of long-range dependencies while maintaining computational tractability. Longformer and related architectures employ sparse attention mechanisms to reduce computational complexity while retaining the ability to handle extended sequences [11, 15].

Recent advances have pushed sequence length capabilities considerably further. Ring Attention [16] enables the processing of sequences exceeding 500,000 tokens through distributed computation across multiple devices. Infini-attention [12] achieves up to 1 million token contexts through compressive memory mechanisms without proportional increases in memory overhead. However, fundamental challenges persist: KV-cache memory requirements grow linearly with sequence length, and attention mechanism complexity scales quadratically [17].

The FlashAttention algorithm [15] represents a significant optimisation by restructuring the attention computation to avoid full HBM access to the $N \times N$ attention matrix. Standard attention requires HBM accesses of order $\Theta(Nd + N^2)$, while FlashAttention reduces this to $\Theta(N^2 d^2 M^{-1})$ through tiling in on-chip SRAM and staged softmax computation, yielding up to 9× fewer HBM accesses. In practice, FlashAttention achieves a 3× speedup for GPT-2 at 1K sequence length, a 15% performance improvement for BERT-large, and 20× greater memory efficiency, enabling benchmark evaluations on Path-X (16K tokens) and Path-256 (64K tokens) [15]. Ring Attention further addresses inter-device communication latency by overlapping KV-block transfers with attention computation, enabling sequence length to scale linearly with the number of participating devices [16].

5. Implementation Details

5.1 Memory and Bandwidth Analysis

Memory bandwidth increasingly dominates the performance profile of extended-context systems. Whereas conventional systems achieve 60–70% of peak memory bandwidth utilisation, extended-context configurations reach 85–95%, inducing memory saturation characterised by unpredictable latency and throughput degradation that renders such systems unsuitable for production environments. KV-cache scaling further amplifies memory pressure: cache size scales linearly with the number of attention heads and is multiplied by the batch size during batched inference. IO-aware approaches such as FlashAttention provide partial relief through SRAM tiling and selective recomputation during backpropagation, storing only model outputs and softmax statistics rather than the full N×N attention matrix, reducing HBM accesses by up to 9×. In production deployments, these optimisations deliver a 3× speedup for GPT-2, a 15% improvement for BERT-large, and 20× greater memory efficiency, making large-scale benchmarks such as Path-X (16K) and Path-256 (64K) tractable [15].

5.2 Energy Efficiency Analysis

The energy consumption of extended-context models for processing long sequences is between 3 and 5 times greater than that of RAG models operating on equivalent tasks. This disparity arises primarily from memory access patterns: accessing data from DRAM consumes approximately 100 times more energy per bit than accessing from cache memory, and 1,000 times more than register access. Extended-context models require large KV-caches resident in DRAM, resulting in sustained high-energy memory access during inference. By contrast, RAG models require only one-tenth of the energy consumed by extended-context models to process an equivalent volume of one million tokens while achieving comparable task performance levels.

5.3 Infrastructure Scalability

Infrastructure analysis identifies three primary challenges for extended-context model deployment. First, interconnect bandwidth: distributed inference for extended-context models requires up to 5–10 times greater inter-device bandwidth compared to RAG-based alternatives, owing to the need to exchange KV-caches across nodes. Second, power density: deploying 70B-parameter models requires approximately 8 GPUs per rack, consuming 40–60 kW of power — significantly exceeding the conventional infrastructure standard of 20–25 kW per rack. Third, thermal constraints: the resulting heat density requires specialised cooling technologies, increasing costs and restricting geographic deployment flexibility. RAG architectures mitigate these pressures by enabling smaller models to operate on standard hardware, with retrieval distributed across storage and compute components.

Ring Attention addresses the interconnect challenge by overlapping KV-block transfer with blockwise attention computation. For a 32-node A100 cluster with a 7B model, this approach extends the effective context from 128K to 4M tokens. For a TPUv4-1024 cluster with a 3B model, the context scales from 32K to 16.384M tokens while preserving computational efficiency. In reinforcement learning settings, Ring Attention enables simultaneous consideration of 128 paths (≈64,000 tokens), compared to a previous limit of 32 paths. A LLaMA-13B model fine-tuned using Ring Attention achieves state-of-the-art line-retrieval performance at up to 512K tokens, significantly outperforming Claude-2 and GPT-3.5 [16].

5.4 Technical Advances in Memory-Efficient Architectures

Infini-attention achieves a 114× compression ratio relative to Memorizing Transformers, enabling effectively infinite context with bounded memory usage by integrating compressive memory directly into the attention mechanism [12]. This approach addresses the core challenge of quadratic memory growth, which would otherwise require 3 TB of memory to support 500B-parameter models with 2,048-token context windows. The Ring Attention technique demonstrates that distributing blockwise self-attention calculations across multiple devices allows training on sequences exceeding 100 million tokens without approximation, achieving a 512× improvement over the prior state-of-the-art in memory-efficient transformers and providing 16M-token contexts for 3B models [27].

The combination of FlashAttention's per-device IO awareness and Ring Attention's distributed ring topology creates a complementary synergy: FlashAttention optimises per-device HBM accesses from $O(N^2)$ to $O(N^2d^2M^{-1})$, while Ring Attention distributes the remaining KV-cache burden linearly across devices. Together, these technologies provide a hardware–algorithm co-design pathway for extending context capabilities to high-resolution long-form video, comprehensive codebase analysis, and scientific dataset processing, without proportional increases in per-device memory or energy consumption.

6. Testing and Validation

6.1 Static Reasoning Performance

Experimental results confirm clear advantages of extended-context transformers for static reasoning tasks in which all necessary information is provided within the query. A 13B-parameter extended-context transformer achieves performance equivalent to a 70B-parameter model using standard context management, indicating that context expansion can serve as a partial substitute for parameter scaling in certain task categories. However, this performance improvement is accompanied by substantially elevated memory costs. KV-cache requirements grow linearly with context size, imposing significant deployment constraints.

Performance gains are considerable when increasing context size from 4K to 32K tokens, providing sufficient capacity for complex multi-step reasoning. Incremental gains diminish beyond 100K tokens, while memory requirements continue to grow exponentially. At 1 million token context sizes, KV-caches for 70B-parameter models require approximately 150 GB of memory — exceeding the capacity of current commercially available GPUs.

6.2 Dynamic Knowledge Task Performance

RAG architectures demonstrate significant advantages in knowledge-intensive tasks. A 7B-parameter RAG model achieves accuracy within 2% of a 70B-parameter parametric model on knowledge-driven benchmarks, while being 10 times smaller and consuming 90% less memory during inference. RAG models are particularly well-suited to temporal queries requiring up-to-date information, domain-specific knowledge retrieval from specialised databases, long-tail knowledge tasks, and tasks requiring synthesis across multiple sources.

Limitations include reduced performance on tasks requiring deep parametric integration of knowledge and degraded accuracy when retrieval quality is poor. Retrieval overhead is, however, modest in practice: dense retrieval over a corpus of 100 million documents requires 10–20 ms per query, accounting for only 5–10% of total latency for a 7B model. This proportion decreases further for larger models in which token generation latency predominates.

6.3 Hybrid Architecture Performance

Systems combining moderate context extension (32K–100K tokens) with RAG retrieval demonstrate superior overall efficiency. A 13B hybrid configuration (32K context combined with RAG) achieves 95% of the reasoning performance of pure extended-context systems, 98% of the knowledge task performance of pure RAG systems, while consuming only 50% of the memory required by extended-context approaches and achieving 80% greater energy efficiency relative to either individual approach. This configuration fits within a single-GPU deployment, eliminating the complexity of distributed inference while maintaining energy consumption levels compatible with large-scale deployment.

6.4 Quality and Reliability Challenges

Beyond architectural efficiency, LLMs face substantive challenges regarding output quality and reliability. Multiple studies indicate that these models generate erroneous or fabricated citations at rates as high as 60% in medical content contexts [21]. Aljamaan et al. [22] developed the Reference Hallucination Score (RHS) metric, which identified critical citation accuracy failures in models including ChatGPT and Bing. Analysis of approximately 4,900 scientific summaries generated by LLMs — including DeepSeek, ChatGPT-4o, and LLaMA 3.3 70B — found that overgeneralisation errors appeared in 26% to 73% of outputs depending on the model [23]. AI-generated scientific abstracts were more than four times more likely to contain overly general statements than human-authored equivalents (odds ratio = 4.85, $p < 0.001$). These quality issues present particular risks in healthcare education, where inaccuracies may compromise academic integrity and propagate misinformation [24].

7. Major Findings / Outcomes / Output / Results

7.1 Real-World Productivity and Accessibility Benefits

Empirical evidence demonstrates measurable productivity gains from the deployment of LLMs in professional settings. Research on GitHub Copilot indicates that developers using AI assistance complete tasks up to 55.8% faster than non-users (95% CI: 21%–89%) [25]. A randomised controlled trial involving 95 programmers confirmed this effect: participants in the AI-assisted group required 71.17 minutes on average to complete an HTTP server assignment, compared to 160.89 minutes for the control group. Notably, participants in both groups estimated their own productivity improvement at only 35%, indicating that self-reported assessments substantially underestimate actual gains. AI users demonstrated willingness to pay $27.25 per month for the tool, compared to $16.91 among non-users, reflecting perceived value beyond speed alone. The greatest productivity benefits were observed among less experienced developers, those with high daily programming activity, and individuals aged 25–44.

A Passive Exposure study involving 844 participants across 48 countries administered the Alternate Uses Task (AUT) to assess the impact of AI-generated ideas on collective creativity [28]. Exposure to ChatGPT-generated responses produced a significant increase in the diversity of ideas generated collectively (Cliff's delta = 0.31, $p = 0.001$), though no significant improvement in individual idea quality was observed ($p = 0.97$). AI-generated ideas were rated as more original than average human responses (beta = 0.62), and AI assistance reduced the tendency towards ideational convergence that typically emerges in group settings.

LLMs also demonstrate considerable utility for multilingual scientific communication. Models have been shown to translate scientific articles with complex specialised terminology into 28 languages while preserving the original JATS XML structure [26], achieving consistent translation accuracy of 91.7% to 98.0%. Expert reviewers assessed 93.3% of translated papers as accurate, suggesting that LLM-assisted translation could substantially broaden access to scientific literature for non-English-speaking researchers globally.

7.2 Economic Exposure and Labour-Market Implications

The Anthropic Economic Index [29] introduces the concept of Observed Exposure, which measures the practical deployment of LLMs across occupational functions rather than theoretical capability alone. The index weights tasks by their operational significance, assigning greater importance to automated API-based tasks than to conversational interactions. A striking finding is the gap between theoretical and observed automation potential: while approximately 94% of tasks in Computer and Mathematics occupations could theoretically be performed by LLMs, only 33% are observed to be automated in practice — due to factors including infrastructure constraints, regulatory frameworks, and workflow adoption barriers.

Occupations with the highest observed exposure rates include Computer Programmers (75%), Customer Service Representatives (71%), Data Entry Keyers (67%), and Financial Analysts (61%). Individuals employed in highly exposed roles tend to be older, predominantly female, better-educated, and higher-earning compared to those in lower-exposure occupations. Despite high exposure levels, no significant increase in unemployment has been recorded in these categories through early 2026. However, a "hiring chill" is emerging: the number of individuals aged 22–25 entering highly exposed occupations declined by approximately 14% between 2022 and 2024, raising concerns about a future junior talent gap in software engineering and financial services. According to the Bureau of Labor Statistics, each 10% increase in AI penetration is associated with a projected 0.6% decline in employment growth by 2034 [29].

7.3 Implications for Architectural Design

The observed gap between theoretical and practical LLM automation capability has significant implications for architectural investment decisions. The substantial capital expenditure associated with building extended-context LLM infrastructure does not translate proportionally into realised automation at current adoption levels. The 94% vs. 33% observed-to-theoretical ratio for Computer and Mathematics occupations suggests that factors beyond model capability — including infrastructure readiness, regulatory constraints, and organisational workflow integration — are the primary limiting factors for practical automation.

Efficient, distributed architectures such as RAG and hybrid systems reduce resource dependency, enabling a wider range of deployment contexts and mitigating risks of capability and control concentration in resource-intensive, centralised infrastructure. Should real-world exposure rates converge toward theoretical limits without commensurate improvements in system efficiency, the environmental impact of large-scale LLM deployment would increase substantially.

7.4 Environmental and Sustainability Considerations

The energy consumption profiles of LLM architectures carry significant environmental implications. Transitioning to efficient architectural designs — particularly RAG and hybrid approaches — can reduce the carbon emissions of LLM deployment by 50 to 80 percent without commensurate loss of computational performance. Comprehensive sustainability assessment must account for the entire model lifecycle, including energy consumed during hardware manufacturing, training, inference, and decommissioning.

Policy mechanisms including carbon pricing, mandatory energy-use transparency reporting, and sustainable procurement requirements in government contracts are identified as effective instruments for accelerating the transition to environmentally responsible LLM design practices.

7.5 Quality, Reliability, and Systemic Risk

RAG systems demonstrate a structural advantage in terms of factual accuracy, as grounding generation in retrieved documents substantially reduces hallucination rates. Retrieval-based systems such as SciSpace and Elicit exhibit markedly lower rates of fabricated content compared to purely generative models [22]. Nevertheless, even retrieval-augmented systems do not fully resolve challenges such as scientific overgeneralisation, academic integrity risks in educational contexts, and the potential for misinformation propagation [24]. Structural efficiency improvements alone are therefore insufficient to address all risks associated with large-scale LLM deployment.

The concentration of LLM infrastructure in entities with substantial financial resources introduces systemic risks of capability monopolisation and single points of failure. RAG and hybrid architectures, by reducing computational resource requirements, enable broader access and distribute capability across a wider ecosystem. The fault tolerance and distributional properties of efficient architectures contribute to overall system resilience and reduce the risks of critical dependence on any single infrastructure component.

7.6 Economic Sustainability and the Observed-Exposure Gap

The discrepancy between infrastructure investment in LLM systems and the observed rates of practical automation raises fundamental questions about return on investment. Current deployment patterns — in which large capital expenditures on extended-context infrastructure yield only 33% observed automation in theoretically 94%-automatable occupations — indicate that infrastructure, regulatory, and adoption barriers significantly limit the economic returns of LLM investment at present. The 14% decline in junior workforce entry into high-exposure occupations since 2022 suggests that the structural labour-market adjustments associated with AI adoption are already underway, even as practical automation levels remain substantially below theoretical ceilings. Architectural choices that improve efficiency and accessibility can help bridge this gap while limiting the environmental costs of any increase in real-world exposure.

8. Conclusion and Future Scope

8.1 Conclusion

This report has examined the strengths and limitations of contemporary large language model architectures in the context of real-world deployment. The Token Economics framework provides a principled basis for comparing extended-context and retrieval-based architectures across dimensions of efficiency, scalability, and sustainability. The analysis reveals that meaningful trade-offs must be understood and managed in order to enable sustainable progress in the field.

Extended-context models are effective for tasks requiring persistent in-context concentration on large bodies of information. However, they incur substantially higher memory usage, exhibit rapid performance degradation at very long sequence lengths, and consume 3 to 5 times more energy per token than retrieval-based alternatives. Retrieval-augmented models, by contrast, excel in dynamic knowledge tasks and achieve up to 80% lower environmental impact relative to extended-context counterparts. The evidence indicates that hybrid architectures, combining moderate context extension (32K–100K tokens) with RAG retrieval, offer the most favourable balance of reasoning capability, knowledge access, memory efficiency, and energy consumption.

Beyond architectural considerations, the report documents significant real-world benefits of LLM deployment, including 55.8% productivity improvements in software development [25] and high-accuracy multilingual scientific translation across 28 languages [26]. Novel architectures such as Infini-attention and Ring Attention provide practical pathways for extending context capabilities without proportional memory scaling. The findings suggest that the practical performance ceiling for LLMs may be determined more by underlying hardware infrastructure than by model architecture alone, and that responsible, efficient design is essential to maximising the societal benefits of these technologies.

8.2 Limitations and Future Directions

This study carries certain limitations that should be acknowledged. The experimental scope does not encompass the most recent models with parameters in the trillions, and caution is therefore warranted when extrapolating findings to architectures at this scale. Rapid advances in hardware may reduce the relevance of some current constraints. Additionally, different model architectures and concrete application domains may alter the cost-benefit trade-offs identified in this analysis.

Productive directions for future research include: (1) adaptive architectures capable of dynamically selecting between extended context and retrieval depending on task requirements; (2) co-design of hardware and algorithms to more efficiently support large-scale context processing; (3) comprehensive risk-benefit analyses encompassing the broader societal and labour-market implications of LLM deployment; and (4) longitudinal studies examining the full lifecycle environmental and economic costs of LLM systems across diverse deployment contexts.

References

[1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, “Attention is all you need,” in Advances in Neural Information Processing Systems, vol. 30, 2017.

[2] T. Brown, B. Mann, N. Ryder, M. Subbiah, J. D. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell et al., “Language models are few-shot learners,” Advances in Neural Information Processing Systems, vol. 33, pp. 1877–1901, 2020.

[3] R. Pope, S. Douglas, A. Chowdhery, J. Devlin, J. Bradbury, J. Heek, K. Xiao, S. Agrawal, and J. Dean, “Efficiently scaling transformer inference,” Proceedings of Machine Learning Systems, vol. 5, 2023.

[4] D. A. Patterson, J. Gonzalez, G. E. Suh, A. B. Kahng, and K. Asanovic, “Carbon in the cloud: Lessons from the internet for sustainable AI,” in 2022 IEEE International Symposium on High-Performance Computer Architecture (HPCA), IEEE, 2022, pp. 1–2.

[5] J. W. Ayers, A. Polastro, P. Leal, A. M. Leas, and M. H. Dredze, “Comparing physician and artificial intelligence chatbot responses to patient questions posted to a public social media forum,” JAMA Internal Medicine, vol. 183, no. 6, pp. 589–596, 2023.

[6] K. Kasneci, K. Sessler, S. Küchemann, M. Bannert, D. Dementieva, F. Gassner, U. Stannat, A. Pfeiffer, I. Vorobev, and E. Aydin et al., “ChatGPT for good? On opportunities and challenges of large language models for education,” Learning and Individual Differences, vol. 103, p. 102274, 2023.

[7] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. Yih, T. Rocktäschel et al., “Retrieval-augmented generation for knowledge-intensive NLP tasks,” Advances in Neural Information Processing Systems, vol. 33, pp. 9459–9474, 2020.

[8] W. Xiong, J. Liu, I. Molybog, H. Zhang, P. Bhargava, R. Hou, L. Martin, R. Rungta, K. A. Sankararaman, and B. Oguz et al., “Effective long-context scaling of foundation models,” arXiv preprint arXiv:2309.16039, 2023.

[9] J. Kaplan, S. McCandlish, T. Henighan, T. B. Brown, B. Chess, S. Sidor, U. Gandhi, and D. Amodei, “Scaling laws for neural language models,” arXiv preprint arXiv:2001.08361, 2020.

[10] Z. Dai, Z. Yang, Y. Yang, J. Carbonell, Q. V. Le, and R. Salakhutdinov, “Transformer-XL: Attentive language models beyond a fixed-length context,” arXiv preprint arXiv:1901.02860, 2019.

[11] I. Beltagy, M. E. Peters, and A. Cohan, “Longformer: The long-document transformer,” arXiv preprint arXiv:2004.05150, 2020.

[12] T. Munkhdalai, M. Faruqui, and S. Gopal, “Leave no context behind: Efficient infinite context transformers with Infini-attention,” arXiv preprint arXiv:2404.07143, 2024.

[13] K. Guu, K. Lee, K. Tung, D. Pasupat, and M. Chang, “Retrieval augmented language model pre-training,” in International Conference on Machine Learning, PMLR, 2020, pp. 3929–3938.

[14] P. Lewis et al., “Retrieval-augmented generation for knowledge-intensive NLP tasks,” Advances in Neural Information Processing Systems, vol. 33, pp. 9459–9474, 2020.

[15] T. Dao, D. Fu, S. Ermon, A. Rudra, and C. Ré, “FlashAttention: Fast and memory-efficient exact attention with IO-awareness,” Advances in Neural Information Processing Systems, vol. 35, pp. 16344–16359, 2022.

[16] H. Liu, M. Zaharia, and P. Abbeel, “Ring attention with blockwise transformers for near-infinite context,” arXiv preprint arXiv:2310.01889, 2023.

[17] N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, and P. Liang, “Lost in the middle: How language models use long contexts,” Transactions of the Association for Computational Linguistics, vol. 12, pp. 157–173, 2024.

[18] R. Pope, S. Douglas, A. Chowdhery, J. Devlin, J. Bradbury, J. Heek, K. Xiao, S. Agrawal, and J. Dean, “Efficiently scaling transformer inference,” Proceedings of Machine Learning Systems, vol. 5, 2023.

[19] D. Patterson, J. Gonzalez, Q. Le, C. Liang, L. Munguia, D. Rothchild, D. So, M. Texier, and J. Dean, “Carbon emissions and large neural network training,” arXiv preprint arXiv:2104.02203, 2021.

[20] M. Nagel, R. A. Amjad, and C. H. Teuscher, “A white paper on neural network quantization,” arXiv preprint arXiv:2103.13622, 2021.

[21] M. Bhattacharyya, V. M. Miller, D. Bhattacharyya, and L. E. Miller, “High rates of fabricated and inaccurate references in ChatGPT-generated medical content,” Cureus, vol. 15, no. 7, p. e39238, 2023.

[22] F. Aljamaan, M. Temsah, I. Altamimi, A. Al-Eyadhy, A. Jamal, K. Alhasan, T. A. Mesallam, and M. Farahat, “Reference hallucination score for medical artificial intelligence chatbots: Development and usability study,” JMIR Medical Informatics, vol. 12, p. e54345, 2024.

[23] U. Peters and B. Chin-Yee, “Large language models as science communicators: A systematic analysis of problematic generalizations in chatbot science summaries,” Proceedings of the Royal Society A, 2025.

[24] J. Kim and B. N. Vajravelu, “Large language models in health care education: A comprehensive review of risks and limitations,” JMIR Formative Research, vol. 9, p. e51319, 2025.

[25] S. Peng, E. Kalliamvakou, P. Cihon, and M. Demirer, “The impact of AI on developer productivity: Evidence from GitHub Copilot,” arXiv preprint arXiv:2302.06590, 2023.

[26] H. C. Kleidermacher and J. Zou, “Science across languages: Assessing LLM multilingual translation of scientific papers,” arXiv preprint arXiv:2502.17882, 2025.

[27] H. Liu, M. Zaharia, and P. Abbeel, “Ring attention with blockwise transformers for near-infinite context,” arXiv preprint arXiv:2310.01889, 2023.

[28] B. Sarkar, R. Dhaliwal et al., “AI and collective idea diversity: Evidence from a global dynamic Alternate Uses Task study with 844 participants across 48 countries,” 2024.

[29] Anthropic, “The Anthropic Economic Index: Observed vs. theoretical AI exposure across occupations,” Anthropic, 2025.